Macroeconomics III / Assignment 1

Consumption and Saving

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1 MPC with infinite horizon

a) Deriving the Euler equation

a.1 Bellman equation

I solve this problem using dynamic programming. You could also use Lagrange optimization as this is a static problem (everything is known at period $t = 0$) - this is done in section 2.

The (representative) consumer faces the following utility maximization problem

$$\max_{c_t, a_{t+1} \forall t} U_t \equiv \sum_{k=0}^{\infty} \beta^{t+k} \frac{c_{t+k}^{1-\sigma}}{1-\sigma}$$

$$s.t.$$

$$c_t + a_{t+1} = y_t + (1+r)a_t$$

$$\lim_{T \to \infty} \left(\prod_{t=0}^{T} (1+r)^{-1} \right) a_{T+1} \geq 0,$$

where a_0 is given and $\sigma > 1$. First I derive the Bellman equation explicitly (for an arbitrary starting index t, which will usually be set to $t = 0$)

$$\begin{aligned}
V_t(a_t) &\equiv \max_{\{c_{t+k}, a_{t+1+k}\}_{k=0}^{\infty}} \underbrace{\left\{ \sum_{k=0}^{\infty} \beta^k \frac{c_{t+k}^{1-\sigma}}{1-\sigma} \right\}}_{U_t} \\
&= \max_{\{c_{t+k}, a_{t+1+k}\}_{k=0}^{\infty}} \left\{ \frac{c_t^{1-\sigma}}{1-\sigma} + \beta \sum_{k=0}^{\infty} \beta^k \frac{c_{t+k+1}^{1-\sigma}}{1-\sigma} \right\} \\
&= \max_{c_t, a_{t+1}} \left\{ \frac{c_t^{1-\sigma}}{1-\sigma} + \beta V_{t+1}(a_{t+1}) \right\} \\
&= \max_{a_{t+1}} \left\{ \frac{(y_t + (1+r)a_t - a_{t+1})^{1-\sigma}}{1-\sigma} + \beta V_{t+1}(a_{t+1}) \right\}.
\end{aligned}$$

The first order condition for an internal optima (for the right hand side) is

given by

$$\underbrace{(y_t + (1+r)a_t - a_{t+1})^{-\sigma}}_{u'(c_t)} = \beta V'_{t+1}(a_{t+1}), \tag{1.1}$$

where we have defined $V'_{t+1}(a_{t+1}) \equiv \frac{\partial V_{t+1}}{\partial a_{t+1}}(a_{t+1})$. By the envelope theorem 1.1 (assuming the value function V_t is differentiable in a_t for all t) we have

$$V'_t(a_t) = (1+r)\underbrace{c_t^{-\sigma}}_{u'(c_t)}. \tag{1.2}$$

Theorem 1.1 (The envelope theorem)**.** *Suppose the household's optimal saving choice is interior and its policy and value functions are differentiable. Then the marginal value of assets is*

$$V'_t(a_t) = (1+r)u'(c_t),$$

For CRRA utility, this becomes

$$V'_t(a_t) = (1+r)c_t^{-\sigma}.$$

Thus, the marginal value of assets equals the additional resources they generate, $1+r$*, multiplied by the marginal utility of current consumption.*

Proof. Define the optimal asset policy $a^*_{t+1} = g_t(a_t)$ as the function that solves the Bellman equation from earlier:

$$V_t(a_t) = u\left(y_t + (1+r)a_t - g_t(a_t)\right) + \beta V_{t+1}\left(g_t(a_t)\right).$$

Differentiating V_t w.r.t. a_t yields

$$\begin{aligned} V'_t(a_t) = & \underbrace{u'(c_t)\left((1+r) - g'_t(a_t)\right)}_{\text{gain due to increased current consumption}} + \underbrace{\beta V'_{t+1}\left(g(a_t)\right)g'_t(a_t)}_{\text{gain due to increased savings}} \\ & = (1+r)u'(c_t) + g'_t(a)\underbrace{\left(\beta V'_{t+1}\left(g(a_t)\right) - u'(c_t)\right)}_{=0 \text{ (because of FOC)}}. \\ & = (1+r)y'(c_t). \end{aligned}$$

□

Intuitively, $\beta V'_{t+1}(g_t(a_t)) - u'(c_t) = 0$ because the marginal gain from current spending equals the discounted marginal gain from saving at the optimum. Therefore, shifting a little between the two has no first-order effect on the value function.

An increase in assets provides additional resources that are distributed between current consumption and savings (and thus future consumption). To first order, this distribution does not matter because both uses have the same marginal value at the optimum. One extra unit of assets provides $1 + r$ extra units of resources, each worth $u'(c_t)$ at the margin, so the increase in value is $(1 + r)u'(c_t)$.

a.2 The Euler equation

Combining (1.1) and (1.2) we obtain the well known Euler equation

$$c_t^{-\sigma} = \beta(1 + r)c_{t+1}^{-\sigma}, \tag{1.3}$$

which is a necessary condition for an internal optimal consumption path. This has a very nice interpretation: The marginal utility today (e.g. $c_t^{-\sigma}$) should equal the marginal utility of saving a little bit and spending it tomorrow (e.g. $\beta(1 + r)c_{t+1}^{-\sigma}$) from todays perspective. If that was not the case then the household could shift consumption and increase total utility.

Equation (1.3) also implies three cases for the consumption path. (i) if $\beta(1 + r) > 0$ then consumption will increase:

$$\left(\frac{c_{t+1}}{c_t}\right)^{\sigma} = \beta(1 + r) > 1,$$

remembering that $\sigma > 1$ is positive. Here the household faces a choice each period of spending a little less but gaining more from increased future consumption through the rent (e.g. $1 + r$) compared to their discount rate (e.g. β). For $\beta(1 + r) < 1$ the story is reversed (limiting consumption today will lead to less discounted future cash) so the consumption path is

decreasing. The case of constant consumption (e.g. $\beta(1+r)$) is discussed in section 1.c.

b) The budget constraint

b.1 Transversality condition

From the maximization problem we have the no ponzi-scheme condition

$$\lim_{T\to\infty}\left(\prod_{t=0}^{T}(1+r)^{-1}\right)a_{T+1}=\lim_{T\to\infty}\left((1+r)^{-(T+1)}\right)a_{T+1}\geq 0,$$

which is a constraint imposed on the consumer to avoid the unrealistic case of infinite debt accumulation.

Because the utility function is monotone in consumption

$$\frac{\partial U_t}{\partial c_{t+k}}>0,\quad \forall k\geq 0,$$

and the consumer gains no direct utility from assets

$$\frac{\partial U_t}{\partial a_{t+k}}=0,\quad \forall k\geq 0,$$

we conclude that the consumer would not increase savings indefinitely, and such the transversality condition is born

$$\lim_{T\to\infty}\left((1+r)^{-(T+1)}\right)a_{T+1}=0. \tag{1.4}$$

b.2 Budget constraint

Iterating forward the budget constraint yields

$$\begin{aligned}
a_t &= (1+r)^{-1}(c_t - y_t) + (1+r)^{-1}a_{t+1} \\
&= (1+r)^{-1}(c_t - y_t) + (1+r)^{-1}\left((1+r)^{-1}(c_{t+1} - y_{t+1}) + (1+r)^{-1}a_{t+2}\right) \\
&= (1+r)^{-1}(c_t - y_t) + (1+r)^{-2}(c_{t+1} - y_{t+1}) + (1+r)^{-2}a_{t+2} \\
&= (1+r)^{-1}(c_t - y_t) + (1+r)^{-1} + (1+r)^{-2}(c_{t+1} - y_{t+1}) \\
&\qquad + (1+r)^{-3}(c_{t+2} - y_{t+2}) + (1+r)^{-3}a_{t+3} \\
&= \sum_{k=0}^{T}(1+r)^{-(k+1)}(c_{t+k} - y_{t+k}) + (1+r)^{-(T+1)}a_{T+1}.
\end{aligned}$$

Taking the limit $T \to \infty$, evaluating at the starting period $t = 0$ and using (1.4) yields

$$\begin{aligned}
a_0 &= \sum_{k=0}^{\infty}(1+r)^{-(k+1)}(c_k - y_k) + \lim_{T\to\infty}(1+r)^{-(T+1)}a_{T+1} \\
&= \sum_{k=0}^{\infty}(1+r)^{-(k+1)}(c_k - y_k) \iff \\
&\qquad \sum_{k=0}^{\infty}(1+r)^{-k}c_k = (1+r)a_0 + h_0, \qquad (1.5)
\end{aligned}$$

where $h_0 \equiv \sum_{k=0}^{\infty}(1+r)^{-k}y_k$ is the present value of human capital / income stream. That is the present value of consumption is equal to the value of period-0 assets and the present value of income.

c) Consumption function

Firstly we rewrite the Euler equation (1.3)

$$c_t^{-\sigma} = \beta(1+r)c_{t+1}^{-\sigma} \iff c_t = (\beta(1+r))^{-1/\sigma}c_{t+1},$$

and derive the consumption at every period in terms of initial consumption

c_0:

$$c_t = (\beta(1+r))^{1/\sigma} c_{t-1} = (\beta(1+r))^{t/\sigma} c_0. \tag{1.6}$$

Substituting (1.6) for all k into the budget constraint (1.5) yields the consumption function

$$\left[\sum_{k=0}^{\infty} \left(\frac{(\beta(1+r)^{1/\sigma}}{1+r}\right)^k\right] c_0 = \left(1 - \frac{(\beta(1+r)^{1/\sigma}}{1+r}\right)^{-1} c_0 = (1+r)a_0 + h_0 \iff$$

$$c_0 = \left(1 - \frac{(\beta(1+r)^{1/\sigma}}{1+r}\right) ((1+r)a_0 + h_0)\,. \tag{1.7}$$

Equation (1.7) is the consumption function describing how consumption today (e.g. period 0) depends on current assets (e.g. $(1+r)a_0$) and the total discounted income stream (e.g. $h_0 \equiv \sum_{k=0}^{\infty}(1+r)^{-k}y_k$).

For $\beta(1+r) = 1$ the Euler equation (1.3) implies

$$c_t^{-\sigma} = c_{t+1}^{-\sigma} \iff c_t = c_{t+1}.$$

That is, stable consumption across all periods. When $\beta(1+r) = 1$ we have that discounting due to delayed consumption is exactly offset by increased consumption in the next period due to the rent received by saving. I think the assignment wants me to answer PIH (the permanent income hypothesis) but I do not agree. PIH refers to households making decisions based on their total future discounted cash-flows. This is captured in (1.5) and (1.7) but does not directly imply constant consumption - as far as my understanding goes.

d) Marginal Propensities to Consume (MPCs)

Using (1.7) we calculate the MPCs of the model. Firstly the MPC of a *one-of* or *windfall* income increase in any given period $t = k$:

$$\frac{\partial c_0}{\partial y_k} = \left(1 - \frac{(\beta(1+r)^{1/\sigma}}{1+r}\right)\left(\frac{1}{1+r}\right)^k, \tag{1.8}$$

and secondly a permanent income increase (i.e. where income in each period increases by the same amount)

$$\frac{\partial c_0}{\partial \Delta y} \equiv \sum_{k=0}^{\infty} \frac{\partial c_0}{\partial y_k} = \left(1 - \frac{(\beta(1+r)^{1/\sigma}}{1+r}\right)\frac{1}{1-(1+r)^{-1}} = \left(1 - \frac{(\beta(1+r)^{1/\sigma}}{1+r}\right)\frac{1+r}{r}. \tag{1.9}$$

Note that for $\beta(1+r) = 1$ where the Euler equation (1.3) implies that consumption is constant (as we saw before) we have $\frac{\partial c_0}{\partial \Delta y} = 1$.

Using the provided calibration we obtain

$$\frac{\partial c_0}{\partial y_0} = 1 - \frac{(0.95(1+0.04)^{1/2}}{1+0.04} \approx 0.044,$$

$$\frac{\partial c_0}{\partial \Delta y} = \left(1 - \frac{(0.95(1+0.04)^{1/2}}{1+0.04}\right)\frac{1+0.04}{0.04} \approx 1.150.$$

The resulting MPCs are lower than any conventional empirical estimate and therefore the model is plausibly wrong in some fundamental way. One very economics coated argument could be that people are credit constrained in the real world. This would make the MPCs for credit constrained people larger because they are below their unconstrained optimal consumption path. Another classic explanation has to do with mortality risk: people are always at risk of dying and if they do not fully care about their bequest as strongly as their own lived life, then MPC out of current income will rise. Simply because future states are discounted more heavily.

My own speculation would be that people do not maximize in this way where they take the whole future path into account and just consume for

the fun of it. Evidence of this can be found in credit card transaction data where there are HUGE spikes around pay-day which would be hard to fit in a framework where households have diminishing marginal utility.

2 The Ramsey model

a) Firms

The firm maximizes profit in each period (non-dynamically) subject to a Cobb-Douglas production technology. Formally it solves:

$$\max_{K_t, L_t} \left\{ \pi(K_t, L_t) = A K_t^{\alpha} L_t^{1-\alpha} - w_t L_t - r_t K_t \right\}, \tag{2.1}$$

$\alpha \in (0,1)$. The problem (2.1) has the following necessary first order conditions for an internal optimum

$$\frac{\partial \pi}{\partial K_t} = A\alpha K_t^{\alpha-1} L_t^{1-\alpha} - r_t = 0 \iff \alpha A k_t^{\alpha-1} = r_t, \tag{2.2}$$

$$\frac{\partial \pi}{\partial L_t} = A K_t^{\alpha} (1-\alpha) L_t^{-\alpha} - w_t = 0 \iff (1-\alpha) A k_t^{\alpha} = w_t. \tag{2.3}$$

The labor-income share is then

$$\frac{w_t L_t}{A K_t^{\alpha} L_t^{1-\alpha}} = \frac{w_t}{A k_t^{\alpha}} = \frac{(1-\alpha) A k_t^{\alpha}}{A k_t^{\alpha}} = 1 - \alpha. \tag{2.4}$$

b) Empirical labor-income

The labour-income share for Poland and Belgium for the period 1995-2021 is graphed in figure 1. The average for Belgium is 0.564 and for Poland 0.443. In the considered model this pins down α through (2.4) so that

$$0.564 = 1 - \alpha^{BE} \iff \alpha^{BE} = 0.436,$$

$$0.443 = 1 - \alpha^{PL} \iff \alpha^{PL} = 0.557.$$

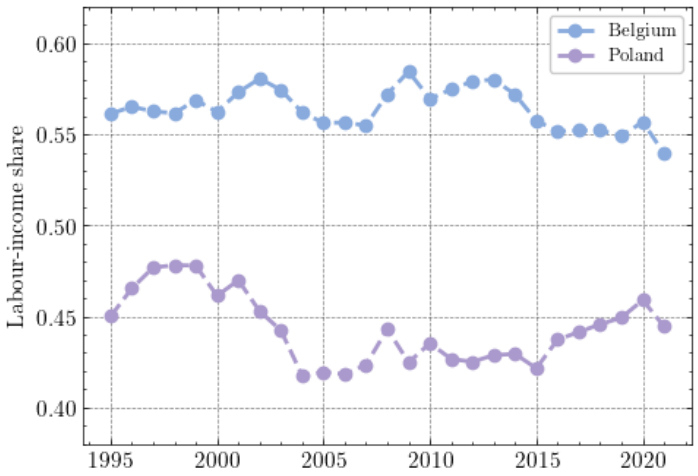


Figure 1: Labour-income share for Poland and Belgium

Note: The labour-income share is calculated as compensation of employees divided by gross value added, both measured at current prices.
Source: National accounts (COMP and VA_CP); own calculations.

c) Households

Households solve the following utility maximization problem

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} U = \sum_{t=0}^{\infty} \beta^t (1+n)^t \frac{c_t^{1-\sigma}}{1-\sigma}$$

$$s.t.$$

$$(1+n)a_{t+1} = a_t R_t + w_t - c_t$$

where $\sigma > 0$, $\sigma \neq 0$, $R_t \equiv 1 + r_t - \delta$ and $a_0 = k_0 > 0$ given. The problem has the following Lagrangian

$$\mathcal{L} = \left[\sum_{t=0}^{\infty} \beta^t (1+n)^t \frac{c_t^{1-\sigma}}{1-\sigma} + \lambda_t \left(a_t R_t + w_t - c_t - (1+n)a_{t+1} \right) \right],$$

with the first order conditions

$$\frac{\partial \mathcal{L}}{\partial c_t} = \beta^t (1+n)^t c_t^{-\sigma} - \lambda_t = 0 \iff \beta^t (1+n)^t c_t^{-\sigma} = \lambda_t,$$

$$\frac{\partial \mathcal{L}}{\partial a_{t+1}} = -(1+n)\lambda_t + \lambda_{t+1}R_{t+1} = 0 \iff \frac{1}{1+n}\lambda_{t+1}R_{t+1} = \lambda_t,$$

and the (somewhat trivial) intertemporal budget constraints, which are found by differentiation wrt. λ_t. Substituting the multipliers for $\lambda_t = \beta^t(1+n)^t c_t^{-\sigma}$ and $\lambda_{t+1} = \beta^{t+1}(1+n)^{t+1}c_{t+1}^{-\sigma}$ yields the consumption Euler equation

$$\frac{1}{1+n}\left(\beta^{t+1}(1+n)^{t+1}c_{t+1}^{-\sigma}\right)R_{t+1} = \beta^t(1+n)^t c_t^{-\sigma} \iff$$

$$c_t^{-\sigma} = \beta \underbrace{(1+r_{t+1}-\delta)}_{R_{t+1}} c_{t+1}^{-\sigma}. \tag{2.5}$$

The interpretation of (2.5) very closely mirrors that of (1.3) with one distinction. Now when households consider shifting consumption to future periods they must also take into account deterioration of their invested capital (e.g. δ). Everything else is identical.

d) Steady state concentration of capital

We impose the asset-market clearing condition

$$a_t = k_t \equiv \frac{K_t}{L_t}, \tag{2.6}$$

for all $t \in \{0, 1, \dots\}$. Using(2.6) along with the steady state (SS) condition $x_t = x_{t+1} = x^{ss}$ for $x \in c, r, k$, the household Euler equation (2.5) and the FOC for firm profit optimization (2.2) we obtain steady state capital per hours worked:

$$\begin{aligned}
(c^{ss})^{-\sigma} &= \beta(1+r^{ss}-\delta)(c^{ss})^{-\sigma} \\
&= \beta(1+\alpha A\,(k^{ss})^{\alpha-1}-\delta)(c^{ss})^{-\sigma} \iff \\
\alpha A\,(k^{ss})^{\alpha-1} &= \frac{1}{\beta}+\delta-1 \iff \\
k^{ss} &= \left(\frac{\alpha A}{\frac{1}{\beta}+\delta-1}\right)^{\frac{1}{1-\alpha}}.
\end{aligned}$$

Interpreting this we see that higher a depreciation rate (e.g. δ) implies a lower k^{ss}. Intuitively this is because investments in capital becomes less attractive (less returns per period) and therefore the balance between consumption and savings (e.g. investments) shifts implying lower k^{ss}.

From here we use that $y_t \equiv \frac{Y_t}{L_t} = \frac{AK_t^{\alpha} L_t^{1-\alpha}}{L_t} = Ak_t^{\alpha}$ and calculate the income capital ratio:

$$z^{ss} \equiv \frac{K^{ss}}{Y^{ss}} = \frac{k^{ss}}{y^{ss}} = \frac{k^{ss}}{A\left(k^{ss}\right)^{\alpha}} = \left(\left(\frac{\alpha A}{\frac{1}{\beta}+\delta-1}\right)^{\frac{1}{1-\alpha}}\right)^{1-\alpha} A^{-1} = \frac{\alpha}{\frac{1}{\beta}+\delta-1}. \tag{2.7}$$

Higher depreciation (e.g. δ) increases the steady state capital-output ratio (e.g. z^{ss}). This is because steady state capital (e.g. k^{ss}) decreases for higher depreciation and because of diminishing marginal returns to capital, this lower lever of capital produce more output on average. Thus z^{ss} decreases in δ.

e) Capital-output ratio

The average capital-output ratio is $z^{BE} = 3.410$ for Belgium and $z^{PL} = 1.576$ for Poland. Assuming both economies are in SS, we consider two cases. Firstly, if depreciation rates are identical, we obtain

$$\frac{\beta^{BE}}{\beta^{PL}} = \frac{1-\delta+0.557/1.576}{1-\delta+0.436/3.410} > 1.$$

That is, Belgian households must be more patient and therefore more willing to postpone consumption. For a common annual depreciation rate of $\delta = 0.05$, we obtain

$$\beta^{BE} = \frac{1}{1-0.05+0.436/3.410} \approx 0.928, \qquad \beta^{PL} = \frac{1}{1-0.05+0.557/1.576} \approx 0.767.$$

Secondly, if discount factors are identical, we obtain

$$\delta^{PL} - \delta^{BE} = \frac{0.557}{1.576} - \frac{0.436}{3.410} \approx 0.226.$$

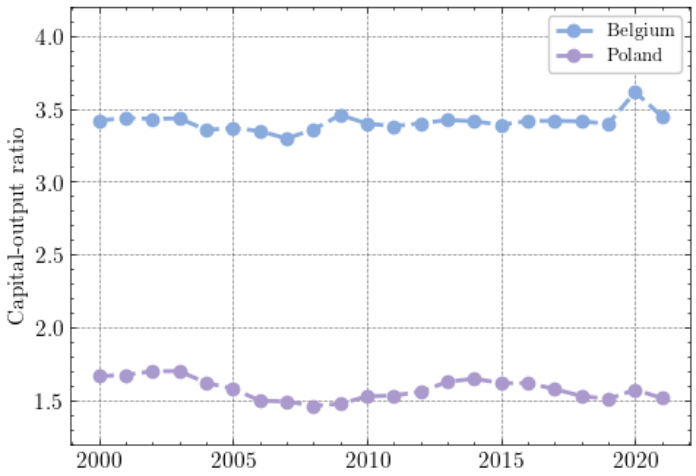


Figure 2: Capital output ratios for Poland and Belgium

Note: The capital-output ratio is calculated as net capital stock divided by gross value added, both measured in chain-linked volumes.
Source: Capital accounts (Kq_GFCF) and national accounts (VA_Q); own calculations.

Poland's annual depreciation rate must therefore be approximately 22.6 percentage points higher. These are substantial differences requiring further justification, although differences in capital composition could explain part of the depreciation gap. Note also that the model assumes closed economies in SS, so these implications need not hold in a richer open-economy framework.